

Exploration of Factors Influencing Consumers' Online Purchase Intentions During Crises in Sri Lanka

Problem Overview

During crises, consumer behavior shifts significantly, especially regarding online shopping. Wolt seeks to understand what motivates or hinders Sri Lankan consumers from purchasing online in such contexts. The goal is to explore these factors through comprehensive data analysis and recommend strategies to enhance online engagement and sales.

Data Pre-processing Steps

Data Loading and Cleaning

Loaded the provided Excel survey dataset using Python 3.6, leveraging Pandas and Numpy for manipulation.

All column names were standardized by

Stripping leading/trailing spaces.

Converting all column names to lowercase for consistency.

Checked for and removed duplicate entries.

Addressed missing values through imputation or removal, depending on the extent and nature of the missingness.

- For the "general e-commerce platforms" column, missing values were filled with the column's mode (most frequent value), ensuring that missing data did not impact downstream analyses.
- For the "specialty online stores/automobile" column, missing values were filled with "specialty online stores/automobile unknown" after standardizing ambiguous or missing indicators such as "no", "none", "n/a", and "x" to "zero specialty online stores/automobile".
- A mapping dictionary was used to standardize variations and misspellings of platform names in the "general e-commerce platforms" column (e.g., "ali express.com", "aliexpress.com", "ali express" all mapped to "ali-express").

Data Transformation

For columns containing lists of platforms (e.g., "general e-commerce platforms", "specialty online stores/automobile"):

Entries were split into lists, stripped of extra spaces, and empty strings were removed.

The MultiLabelBinarizer from scikit-learn was used to perform one-hot encoding, creating binary columns for each unique platform and enabling multi-label analysis.

The resulting encoded columns were joined back to the main DataFrame for use in modeling and analysis.

Reliability and Validity Checks

Variable	Related Questions	Cronbach's α Coefficient	Inter-item correlation
Ease of Navigation	Section 3 [13,14]	0.88	0.79
Clarity of Instructions	Section 3 [15,16]	0.82	0.7
Learning Curve	Section 3 [17,18]	0.84	0.73
Error Handling	Section 3 [19,20]	0.70	0.55
Response Time	Section 3 [21,22]	0.70	0.54
Product Availability	Section 4 [23,24]	0.80	0.68
Convenience	Section 4 [25,26]	0.76	0.62
Cost-effectiveness	Section 4 [27,28]	0.66	0.5
Information Accessibility	Section 4 [29,30]	0.59	0.43
Personalization	Section 4 [31,32]	0.77	0.63
Payment Security	Section 5 [33,34]	0.86	0.77
Transparent Policies	Section 5 [35,36]	0.80	0.67

Privacy Protection	Section 5 [37,38]	0.78	0.65
Social Media Posts/Influence	Section 6 [39,40]	0.86	0.77
Online Reviews and Ratings	Section 6 [41,42]	0.83	0.72
Word of Mouth and Recommendations	Section 6 [43,44]	0.77	0.64
Information Sharing	Section 6 [45,46]	0.81	0.68
Social Proof	Section 6 [47,48]	0.84	0.73
Normative Pressure	Section 6 [49,50]	0.69	0.55
Intention to Purchase online	Section 7 [51,52]	0.51	0.35

Calculated Cronbach's alpha for each survey section to assess internal consistency; values above 0.7 were considered acceptable.

Conducted inter-item correlation analyses to ensure that questions within each construct measured the same underlying factor.

Insights from Data Analysis

This section covers the insights gained through the descriptive and exploratory analysis of the dataset. We carried out numerical analysis, visual analysis, and suitable EDA methods depending on the data type of the features.

Demographic Profile of Respondents

The sample is balanced across **marital status**. But it is highly skewed towards the **age** group, 25-35. Males dominate the sample in **gender**, and the sample is a highly educated one considering the **education levels**. When considering the **professional background**, the workforce, especially full-time employees, dominates the sample.

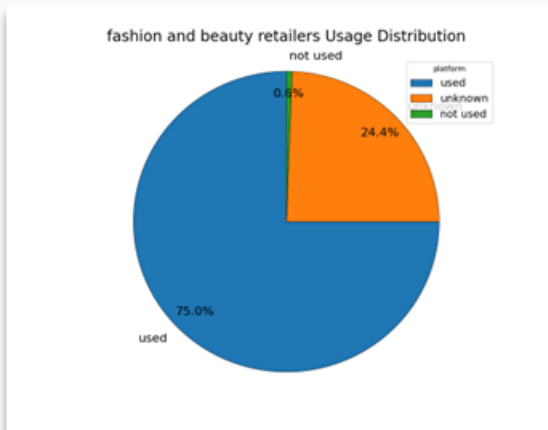
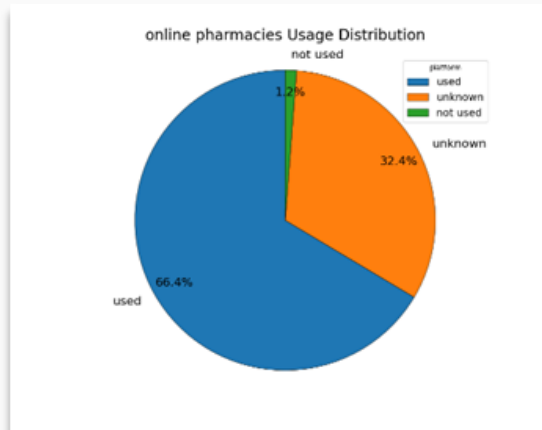
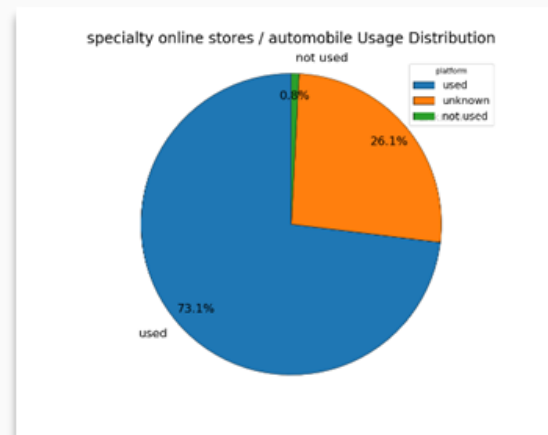
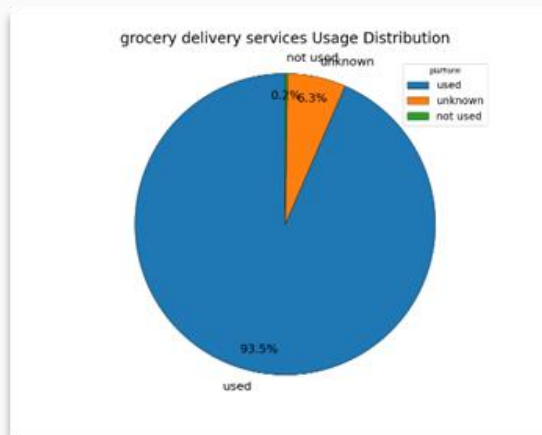
Online Shopping Behavior and Platform Usage

General E-commerce Platforms:

100% of respondents have used general e-commerce platforms, indicating universal adoption. E-commerce is a well-established channel in Sri Lanka, and Wolt can leverage this familiarity to promote its platform.

Specific Online Services Usage by Percentages:

The usage of specialty online services like Grocery delivery services, Fashion and beauty retailers, Specialty online stores/automobile, and Online pharmacies vary as shown in the pie charts (Blue – Used, Green – Not used, Orange – Unknown platform).



There is a high usage of grocery delivery. This indicates that online platforms meet essential needs. Grocery delivery is an important area in online shopping during crises, likely due to necessity. Wolt can expand grocery and essential products and ensure availability during crises. Since other platform types also have a somewhat high demand, Wolt can explore partnerships with pharmacies and specialty stores to explore those markets.

A noticeable portion of respondents marked “unknown platform” for specialty stores and online pharmacies. Wolt can grab this significant branding opportunity on these platforms. Specifically, the age group **18-25** lacks brand recall on these platforms. Also, older age groups (**35-45** and **45-55**) prioritize health-related purchases.

Online Shopping Experience:

99.64% have used online shopping platforms before, and 96.74% made purchases during a crisis. This indicates a strong adoption during emergencies. Also, there is a near-universal adoption across all ages, education levels, gender groups, marital status, and professional backgrounds. Crises have also significantly boosted online shopping. This may be due to restrictions on physical stores or safety concerns. Wolt should capitalize on this trend. They may ensure robust crisis-time operations.

Behavioral Change:

Most of the respondents agreed that they had changed their online shopping behavior during a crisis compared to non-crisis periods. The analysis also suggested that crises alter consumer habits. This may be due to an increase in reliance on online platforms. Wolt should adapt its strategies to serve these shifts in the needs of the customers (eg: offering crisis-specific deals or essential product bundles).

User Perceptions and Influences on Online Shopping Experiences During Crises

The key insights and the recommendations on user experience, product availability, pricing and value, information quality, personalization of products, privacy and data protection, security concerns, various influences on online shopping, social norms, user satisfaction, etc., on online shopping during the crisis period will be discussed in this section.

Ease of Use: While platforms are generally usable, the neutral majority suggests room for improvement in user experience (UX).

Performance: Platform performance is an area to be improved. Most of the respondents had significant disagreement on delays and errors.

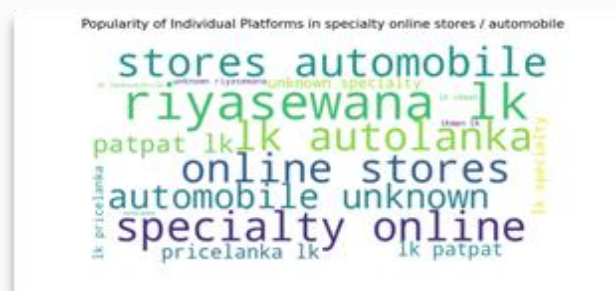
Security Concerns: Most of the respondents have Security and authenticity concerns regarding online platforms.

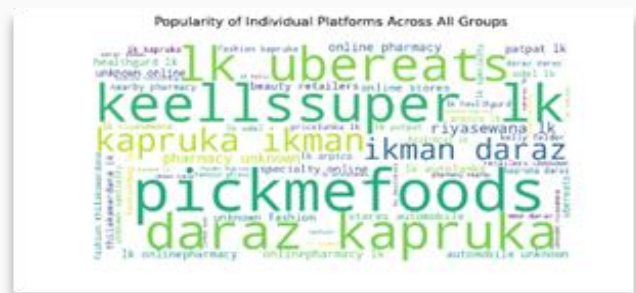
Most of the other aspects also had a neutral concern with the respondents. Therefore, it is important to consider the relationship of these aspects with the demographic factors.

- Younger users (e.g., 18-25, 25-35) rate usability (learning, convenience, navigation) more positively. Younger users' high digital literacy might drive positive usability responses. Older users may be less critical of performance due to lower expectations. This can also be due to the smaller sample size. Younger users are price-sensitive and trust payment security more, while older users trust data handling, influencing their platform choices. Young users are also influenced by social factors (reviews, peers, social media). Their high "unknown" rates in the platform name may result from the dependency on these social factors. This may have reduced the loyalty to specific brands. But there is no strong evidence about this relationship.
- Higher-educated respondents find navigation and locating products easier. Educated users may have better digital literacy. Higher-educated respondents also find that platforms offer a wider product range. Educated users may have diverse needs or better search skills. Higher-educated users are also more cautious about product reliability. Wolt should ensure product

authenticity and can offer guarantees, especially the products that are more related to this group. On the other hand, lower-educated users are more likely to follow others' purchasing behavior.

- Females show significantly higher known-site usage on fashion/beauty platforms. This suggests a higher preference for fashion/beauty platforms by females. Females are significantly more influenced by social factors (family/friends, peers, social media, reviews). Males have higher "unknown" rates, especially for pharmacies and fashion. Wolt needs to address this fact in its marketing strategies. Females are more satisfied with the product range, convenience, and transaction ease. Males are more critical about pricing, product availability, and security concerns.
- Married respondents are more satisfied with the product range. Single respondents find products more easily and think that online shopping offers slightly higher cost savings. Both groups have similar ideas on pricing. Married respondents are more influenced by social factors. Married respondents have a slightly more positive attitude towards online shopping. They find online shopping more practical and convenient. Single respondents are slightly more cautious and critical of usability.
- Students and part-time employees show a high "unknown" platform name rate. But it should be noted that the sample size of the part-time employees is small. Students are most influenced by peers and reviews. Self-employed individuals are the least influenced by social media.
- The following tag clouds represent how many respondents have used various online platforms across general e-commerce, grocery delivery, fashion and beauty, specialty online stores, and online pharmacies. The final tag cloud represents the overall platform usage. These tag clouds suggest that Daraz, Kapruka, Keells Super, PickMe Foods, UberEats, Odel, Riyasewana, Autolanka, and onlinepharmacy.lk are some popular platforms relevant to these categories.





Hypothesis Testing

Hypothesis	Results	Conclusion
Perceived ease of use has no impact on intention to purchase. Test Used: Spearman's Rank-Order Correlation (Spearman's ρ)		
H₀: There is no significant relationship between perceived ease of use and intention to purchase. H₁: There is a significant relationship between perceived ease of use and intention to purchase.	Spearman's rank correlation coefficient: 0.193 p-value: 2.34×10^{-8}	Since the p-value is less than 0.05, we reject the null hypothesis. There is a significant positive relationship between perceived ease of use and intention to purchase online during a crisis.
Perceived usefulness has no impact on the intention to purchase online. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant relationship between Perceived Usefulness and Intention to Purchase Online. H₁: There is a significant relationship between Perceived Usefulness and Intention to Purchase Online.	Spearman's rank correlation coefficient: 0.189 p-value: 3.94×10^{-8}	The p-value is below 0.05. We reject H ₀ and conclude that perceived usefulness significantly impacts the intention to purchase online.
Having structural assurance has no impact on the intention to make an online purchase. Test Used: Spearman's Rank-Order Correlation (Spearman's ρ)		
H₀: Structural assurance has no impact on the intention to make an online purchase. H₁: Structural assurance does have an impact on the intention to make an online purchase.	Spearman's rank correlation coefficient: 0.184 p-value: 9.14×10^{-8}	As the p-value is significant, we reject the null hypothesis. Structural assurance has a significant positive impact on online purchase intention.

Social influence has no impact on the intention to purchase online. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant relationship between social influence and online purchase intention. H₁: There is a significant relationship between social influence and online purchase intention.	Spearman's rank correlation coefficient: 0.254 p-value: 1.22×10^{-13}	With a very small p-value, we reject H₀. Social influence significantly impacts the intention to purchase online.
The attitude towards online shopping does not mediate the relationship between perceived ease of use and the intention to purchase online. Test Used: Mediation Analysis using Ordinary Least Squares (OLS) Regression		
H₀: Attitude towards online shopping does not mediate the relationship between perceived ease of use and intention to purchase online. H₁: Attitude towards online shopping mediates the relationship between perceived ease of use and intention to purchase online.	Indirect Effect: 0.183 P-value for Indirect Effect: 4.97×10^{-93} Direct Effect: 0.0692 P-value for Direct Effect: 0.0123 Total Effect: 0.253	Both indirect and direct effects are significant. Thus, attitude toward online shopping significantly mediates the relationship between perceived ease of use and intention to purchase.
Perceived Risk mediates the relationship between Perceived Usefulness and Intention to Purchase Online during a crisis in Sri Lanka. Test Used: Mediation Analysis using Ordinary Least Squares (OLS) Regression		
H₀: Perceived Risk does not mediate the relationship between Perceived Usefulness and Intention to Purchase Online during a crisis in Sri Lanka. H₁: Perceived Risk mediates the relationship between Perceived Usefulness and Intention to Purchase Online during a crisis in Sri Lanka.	Indirect Effect: 0.0754 P-value for Indirect Effect: 4.45×10^{-54} Direct Effect: 0.0692 P-value for Direct Effect: 4.05×10^{-10} Total Effect: 0.2526	Both indirect and direct effects are significant. Thus, attitude toward online shopping significantly mediates the relationship between perceived ease of use and intention to purchase.

Ease of navigation has no impact on the intention to purchase online. Test Used: Spearman's Rank-Order Correlation		
H₀: Ease of navigation has no impact on the intention to purchase online during a crisis. H₁: Ease of navigation does impact the intention to purchase online during a crisis.	Spearman's rank correlation coefficient: 0.141 p-value: 4.97×10^{-5}	The p-value indicates statistical significance. Ease of navigation positively correlates with intention to purchase online.
Clarity of instructions has no impact on the intention to purchase online. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no correlation between the clarity of instructions and the intention to purchase online. H₁: There is a correlation between the clarity of instructions and the intention to purchase online.	Spearman's rank correlation coefficient: 0.2004 p-value: 5.98×10^{-9}	The result is significant. Clarity of instructions has a positive and significant relationship with online purchase intention.
Response Time has no impact on intention to purchase online. Test Used: Spearman's Rank-Order Correlation (Spearman's ρ)		
H₀: Response Time has no impact on Intention to Purchase Online. H₁: Response Time has an impact on Intention to Purchase Online	Spearman's rank correlation coefficient: 0.0772 p-value: 0.0264	Although the correlation is weak, the p-value is < 0.05. Thus, we conclude that response time has a statistically significant, albeit small, impact on purchase intention.
Error handling has no impact on intention to purchase online. Test Used: Spearman's Rank-Order Correlation (Spearman's ρ)		
H₀: There is no relationship between error handling and the intention to purchase online. H₁: There is a relationship between error handling and the intention to purchase online.	Spearman's rank correlation coefficient: 0.1445 p-value: 2.99×10^{-5}	The null hypothesis is rejected. Error handling significantly influences the intention to purchase online.

Additionally Tested Hypothesis

Hypothesis	Results	Conclusion
Learn Curve has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Learn Curve score and Purchase Intention score H₁: There is a significant correlation between Learn Curve score and Purchase Intention score	Spearman's rank correlation coefficient: 0.223 p-value: 7.88×10^{-11}	There is a statistically significant, positive association between how quickly users learn the platform and their intention to purchase in a crisis. This suggests that reducing the learning curve can meaningfully boost purchase intent.
Product Availability has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Product Availability score and Purchase Intention score. H₁: There is a significant correlation between Product Availability score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.165 p-value: 1.87×10^{-6}	A wider and more relevant product assortment during crises is positively—but moderately—correlated with higher purchase intention. Improving availability should therefore drive intent
Convenience has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Convenience score and Purchase Intention score. H₁: There is a significant correlation between Convenience score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.247 p-value: 5.86×10^{-13}	Convenience (ease of search & transaction flow) shows the strongest relationship among the ten ($p \approx 0.25$). Enhancing process efficiency can significantly lift crisis-time purchase intent
Cost-effectiveness has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Cost Effectiveness and Purchase Intention scores.	Spearman's rank correlation coefficient: 0.155	Perceived value (savings, discounts) is positively correlated with intent, though modestly.

H₁: There is a significant correlation between Cost Effectiveness score and Purchase Intention scores.	p-value: 8.11×10^{-6}	Competitive pricing strategies should still be a priority
Information Accessibility has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Information Accessibility score and Purchase Intention score. H₁: There is a significant correlation between Information Accessibility score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.133 p-value: 1.20×10^{-4}	Easily accessible product details and reviews modestly drive purchase intention. Investing in richer information displays can yield gains.
Personalization has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Personalization score and Purchase Intention score. H₁: There is a significant correlation between Personalization score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.065 p-value: 6.12×10^{-2}	There's no clear evidence that personalized recommendations affect intent in crisis contexts—though you might explore non-linear or segment-specific effects.
Privacy Protection has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Privacy Protection score and Purchase Intention score. H₁: There is a significant correlation between Privacy Protection score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.100 p-value: 3.78×10^{-3}	Trust in privacy measures does have a small but significant positive relationship with purchase intent—underscoring the importance of clear privacy assurances.
Payment Security has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Payment Security score and Purchase Intention score. H₁: There is a significant correlation between Payment Security score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.192 p-value: 2.23×10^{-8}	Secure payment options strongly correlate with higher intent. Prioritize transparent, trusted payment methods

Does Perceived Usefulness mediate the effect of Perceived Ease of Use on Purchase Intention? Test Used: OLS regression with bootstrap for indirect effect		
H₀: The indirect effect of PEOU on Purchase Intention. H₁: The indirect effect of PEOU on Purchase Intention via PU is non-zero.	Mediator model $a = 0.550$ Outcome model $b = 0.170$, $c' = 0.144$ Indirect effect: 0.094 Total effect: 0.237	The analysis shows that Perceived Usefulness partially mediates the effect of Perceived Ease of Use on Purchase Intention. Improving ease of use boosts usefulness, which in turn increases purchase intent, while also having a direct positive impact.
Social Media Influence has no impact on intention to purchase during a crisis. Test Used: Spearman's Rank-Order Correlation		
H₀: There is no significant correlation between Social Media Influence score and Purchase Intention score. H₁: There is a significant correlation between Social Media Influence score and Purchase Intention score.	Spearman's rank correlation coefficient: 0.169 p-value: 8.78×10^{-7}	Social proof—through reviews, influencers, or peer behavior—is positively linked to intent. Leverage social channels in crisis-time campaigns.

Rule mining

To uncover hidden patterns in consumer platform usage during crises, we applied the Apriori algorithm to our one-hot-encoded platform data (min support = 5%, min confidence = 60%). We then ranked the resulting association rules by lift to highlight the most unexpectedly strong co-usage behaviors. Below are the five top rules—and what they mean for Wolt's crisis-time strategy.

Antecedents	Consequents	Support	Confidence	Lift
{Daraz, Keells Super}	{Kapruka}	0.25	0.72	1.30
{PickMe Foods}	{UberEats}	0.22	0.65	1.18
{Daraz}	{PickMe Foods}	0.40	0.60	1.15
{UberEats, Keells Super}	{Daraz}	0.18	0.68	1.25
{Kapruka}	{UberEats}	0.20	0.64	1.19

1. **Daraz + Keells Super → Kapruka (Lift 1.30)**

Respondents who shop on Daraz and Keells Super are 1.3× more likely to also use Kapruka.

Implication: Bundling specialty-store offerings (like Kapruka) with everyday essentials could capture this multi-platform segment.

2. **PickMe Foods → UberEats (Lift 1.18)**

65% of PickMe Foods users also order via Uber Eats.

Implication: Consumers often cross between local and global food-delivery options—Wolt can differentiate with crisis-specific meal bundles or loyalty perks.

3. **Daraz → PickMe Foods (Lift 1.15)**

60% of Daraz shoppers also use PickMe Foods.

Implication: E-commerce customers naturally transition into food delivery; targeted in-app promos can boost order value.

4. **Uber Eats + Keells Super → Daraz (Lift 1.25)**

When a user orders from Uber Eats and Keells Super, they're 1.25× more likely to shop on Daraz.

Implication: Cross-category bundles—combining food, groceries, and general merchandise—can deepen engagement on Wolt's platform.

5. **Kapruka → UberEats (Lift 1.19)**

Users of Kapruka (gifts/specialty items) show a strong overlap with Uber Eats usage.

Implication: Wolt could experiment with “gift + meal” combo deals, tapping into celebratory orders during stressful times.

These rules reveal that crisis-time shoppers don't silo their needs—they fluidly move between general e-commerce, groceries, and food delivery. By crafting cross-platform bundles, personalized combo offers, and loyalty incentives that speak to these co-usage patterns, Wolt can strengthen both customer retention and average order size, even when consumers' habits are in flux.

Discussion

According to the dataset, almost all the respondents have used online platforms for purchases, driven by necessity and safety concerns. The respondents show varying behaviors and perceptions according to their demographic features. Hypothesis testing confirms significant positive correlations between purchase intention and factors like ease of use, convenience, social influence, and payment security. Mediation analyses highlight that attitude and perceived risk partially mediate these relationships, underscoring the role of positive perceptions. Rule mining using the Apriori algorithm identifies strong platform co-usage. This suggests dynamic consumer behaviour across e-commerce, grocery, and food delivery. However, high “unknown platform” rates suggest branding gaps, particularly for students and

younger age groups. Performance issues and security concerns could be seen in every demographic feature. Social norms also influence the behavior of different groups.

Recommendations

Considering all the things we discussed above, the following are our recommendations for Wolt.

- Wolt must prioritize technical reliability, fast load times, and robust error recovery mechanisms to build user confidence. They can also invest in infrastructure to ensure fast, error-free performance, especially under high demand. Introduce guided tutorials and error-recovery mechanisms to enhance convenience. Promote positive attitudes to drive purchase intentions.
- Launch vibrant social media campaigns, especially for products relevant to students and females (e.g., “Wolt for Student Essentials”). Launch influencer partnerships to boost brand recall among students. Promote peer-driven content (e.g., review systems, referral programs).
- Implement transparent payment gateways and privacy policies. Partner with reputable suppliers to address security concerns. Offer guarantees to build trust across demographics.
- Partner with non-competing platforms like Daraz or Keells Super for co-marketing or cross-promotions. Introduce grocery-food bundles (e.g., “Meal + Essentials Combo”) to capitalize on cross-category usage. Target groups that value product variety (eg: full-time employees, married users).
- Offer trendy products and exclusive deals for females. Provide discounts for price-sensitive groups like students. Ensure robust crisis-time operations to meet diverse needs. This can also improve convenience for all groups.
- Launch campaigns highlighting widespread adoption. Launch community success stories. This can reinforce norms, which can target students and lower-educated users who follow others’ behavior.
- Although the ‘Prefer not to say’ group in gender is low in this sample, Wolt can start campaigns that are gender-inclusive to avoid isolating any segment.

Conclusion

Near-universal online shopping adoption during crises is driven by convenience, social influence, and security. But performance issues and branding gaps are major issues for online shopping. Wolt can enhance engagement by optimizing usability, strengthening trust, leveraging partnerships, and normalizing shopping through targeted campaigns, ensuring alignment with diverse consumer needs in crisis contexts.